

Spec — Desk Modules, Tabs and Fields (Customization)

App	TAVI Desk — https://staging-desk.taviportal.com (tenant NDC-Staging)
Feature	Setup → Customization → Modules and Fields (module list, module detail, Layout Builder, Fields, Field Permissions), Data Types , Capabilities , Organize Tabs , and the top tab bar
Scope	Standard modules Tickets, Contacts, Accounts, Products, Calls, Events, Tasks, Contracts; custom modules; tabs; layouts; fields. Time Entry is out of scope for now (user, 5 Oct 2026)
Feature tickets	NDC-1249 "Modules and Fields" (test-case holder), NDC-1489 "Modules and Fields Rules" (layout/validation rules, record locking, dependency, links, buttons — plan item, Not Started)
Gap ticket already in Plane	NDC-1879 "Customization (modules, fields, layouts): features Zoho Desk has that Desk is missing" — 6 items (Z1–Z6 here). The new gaps in this spec (Z8 onward) are not filed — user asked for no Plane changes
Missing list	docs/desk-modules-fields/desk-modules-fields-missing-list.md
Reference notes	docs/desk-modules-fields/reference/ — zoho-desk-live-inventory.md (live Zoho, every module's fields), zoho-docs-inventory.md (help/API/edition facts with sources), tavi-fields-summary.txt (TAVI fields per module)
Reference product	Zoho Desk — help articles Creating custom modules , Standard modules and fields , Customizing standard modules and the linked layout, field, lookup and API pages; plus the live Zoho Desk org (read-only, 5 Oct 2026)
Written	5 Oct 2026, from bundle <code>index-Cegqr3UG.js</code> (module builder chunk <code>pages-DnrcKUi0.js</code> , locales <code>crm-modules</code> , <code>crm-layout-builder</code> , <code>desk</code>), live read-only API and UI (owner and second admin account)
Status	Draft for review

Feature inventory

Built in Desk (found 5 Oct 2026)

Module list (/settings/modules-and-fields)

- Table: Displayed In Tabs As · Module Name · Last Modified · Status (toggle), search, paging 10/25/50/100, reorder ("Module order saved") — B1–B6
- Tabs **Modules**, **Web Tabs** and **Global Sets** — the last two are "In Development" placeholders — B7
- Buttons **Custom Module**, **Team Module**, **Create New Module** — C1–C14
- Row menu: Layouts, Rename, Fields, Button, Links, Access Control, Workflow Rules, Assignment Rules, Map Dependency Fields, Set Validation Rules, Link/Unlink Teamspace, Delete (which items show depends on module type) — B8
- 9 modules seeded: Tickets, Contacts, Accounts, Products, Calls, Events, Tasks, Contracts, Time Entries. No custom module exists on the tenant today

Create / rename / delete a module

- Create: plural name, singular name, description, **Module Data Storage** Organization / Department (cannot change later), **Module Permission** by profile (Desk Administrator, Desk Supervisor, Desk Agent, Desk Light Agent, Desk Reviewer) — C1–C10
- Rename: singular, plural, API name (with warning "Changing the API name may break existing integrations..."), icon, description — D1–D8
- Delete: only when the module has no records ("...still has records. Delete its records first...") — C12, C13

- Enable / disable a module; "This module has dependencies and cannot be disabled." — B5, B6
- Record visibility per module: Public Read/Write/Delete · Public Read Only · Private — H6–H8

Module detail (/settings/modules-and-fields/{moduleId})

- Desk shows only the tabs **Layouts, Fields, Workflow Rules, Summary**. The client also has Layout Rules, Validation Rules, Record Locking, Map Dependency Fields, Links, Buttons, Module Permission, Tags and Lead Conversion Mapping tabs, but Desk does not show them — E1
- Summary: singular, plural, module key, API name, status, storage, description, created/modified by and at, **Access Control** (All Users / Selected Users) — E2, H9
- Fields: **Field Listing** (Fields, Data Type, Custom Field, Layouts; layout filter All Layouts / one layout / Not on any layout) and **Field Permissions** (Read and Write / Read Only / Don't Show per profile; system fields locked to Read Only) — F1–F6, H1–H5

Layouts and Layout Builder (../{moduleId}/layouts/{layoutId})

- One "Standard" layout per module, marked **Default**; Create New Layout, Clone Layout, Set as default, enable/disable, delete with **Move records to** another layout; default layout cannot be disabled or deleted; layout name ≤ 40 chars and unique — G1–G12
- Three views: **CREATE, QUICK CREATE** (one section only), **DETAIL VIEW**; Preview as Create / Quick Create / Detail View — G13–G15
- Sections: add, rename (English / Arabic), 1 or 2 columns, tab order Left to Right / Top to Bottom, move up/down, delete (fields go to Unused Fields; blocked while the section holds a required field) — G16–G21
- New Fields palette (33 types incl. Address, Auto Number, Camera, File Upload, Formula, Image Upload, Location, Long Integer, Lookup, Map, Multi-Select Lookup, Radio Button, Rich Text, Rollup Summary, Status, Subform, User), Unused Fields, "Custom Fields Left: N" (limit 512 per layout in the client) — I1–I3
- Field menu: Mark/Unmark Required, Mark/Remove Unique, Set Permission, Full/Half width, Edit Properties, Create Layout Rule, Create Validation Rule, Capabilities, Remove from Layout — I4–I10
- Field Properties: GENERAL (label EN/AR/FR ≤ 50, API name — fixed after save, description), VALIDATION (Required, Unique incl. across other modules, Regex with tester, Conditional visibility, Required while visible), PERMISSIONS (Read-only, per-profile Read/Edit — saves immediately), ADVANCED (picklist options with replace-on-remove, currency symbol, auto-fill, auto-number, lookup, formula, subform, rollup, quick-create triggers, subscription trigger) — I11–I40
- Remove field: **Move to Unused Fields** or **Delete Permanently**, with a usage check (Workflow rules, Saved actions, Approval processes, Review processes, Telephony) — I41–I46
- Field Dependency (Show / Hide / Require / Make Optional when a field equals any of values) — I47–I50
- Layout Permission (profiles), Rename Module from the builder, module image on/off, integration-contract warning, unsaved-changes dialogs, Reset, Undo/Redo and keyboard shortcuts, Help drawer — G22–G30

Data Types and Capabilities (/settings/data-types , /settings/capabilities)

- Data type catalogue (35 incl. one custom "TestData Type"), add custom data type on a base type, capability flags per data type — J1–J8

Organize Tabs and the tab bar (/settings/organize-tabs)

- Selected / Unselected modules, drag or Move up/down, Hide/Show, "Always shown" lock, Reset to default, "**Changes apply to you only**" — K1–K10
- Top bar: Tickets, Knowledge Base, Customers, Analytics, Activities, Chat, Community, Social, Contracts, "More modules" with search — K11–K14

Missing in Desk (Group Z)

- Already in NDC-1879: layout rules (Z1), validation rules (Z2), picklist values per layout (Z3), Help Center access per field (Z4), nested and colour-coded picklists (Z5), field encryption / ePHI (Z6). Ticket number format is NDC-1866 item 16 (Z7)
- New (not filed): see Group Z, Z8 onward, and the missing list

Not a gap

- Deleting or disabling a custom module (Zoho cannot do either) — Desk extra
- More than one layout on a custom module, auto-number, user field, rich text, subform, rollup, unique across modules — Desk extras
- Import/export of custom module records (Zoho: unsupported) — not required
- Admin default list columns (removed from NDC-1784 as not a gap)

Not yet confirmed

- Whether renaming a standard module changes the tab bar, breadcrumbs, record pages and Organize Tabs (tab labels come from the `desk_locale` — D3)
- Whether a new custom module appears in the tab bar / More modules (no custom module exists; creating one is a test step — C11)
- Whether a field's data type can be changed after save (I12)
- Department accessibility of a custom module: Zoho docs say it can be chosen, but the live New Module form shows "Accessible To" locked to All Departments — not listed as a gap
- "Add to Other Layouts" and "Preview layouts based on profiles" are in Zoho docs but were not seen live (one layout per department in the reference org) — not listed as gaps
- Department-storage modules: the tenant returns **0 departments** for the owner (`GET /desk/departments` → `[]` , HQ shows "Showing all 0 departments") — §1.4

0 How to run

1. **Order:** §1 preconditions → A → B → K → E → F → H → J (read) → C → D → G → I → J (write) → L → E2E → Z. Take the baseline (§1.3) before A and diff it after the last group.
2. **Serial only:** C, D, G, I, J-write, K-write and every `[serial]` row. Module, layout, field and tab metadata is **tenant-wide** — never run two writers at once, and never edit the seeded standard modules' fields except where a row says so (then restore in the same row).
3. **Parallel allowed:** A, B, E, F, H-read, J-read, L-ui and all API GET rows.
4. **Exclusive:** `[perf]` rows run alone, with nothing else signed in to the account.
5. `[data]` rows record: field/type, input (exact bytes for odd input), what the UI showed, what the API stored (GET after save), and pass/fail per row.
6. **Sessions:** the access token is short-lived and rotates; a second sign-in of the same account can revoke the first (seen 5 Oct 2026: `SESSION_REVOKED / login?reason=revoked`). One script = one sign-in; sign out at the end; use the second admin account for long UI runs. Report token **lengths** only.
7. **Verdicts:** Pass / Fail / Blocked / Inconclusive. A row blocked by §1.4 is **Inconclusive, not Fail**.

1 Preconditions

1.1 Accounts

Role	Account (from docs/Credentials/TAVI DESK Credentials.txt)	Status
Tenant owner (admin)	ndc-staging-owner@taviportal.com	Works; has metadata.modules.manage , metadata.fields.manage_permissions , metadata.module_templates.manage
Second admin	mahmoud.mohamed1@taviportal.com	Works (used for the UI captures)
Agent / Light Agent / Reviewer profile 	one of the Gmail accounts in the same file — profile to be confirmed	Needed for A2–A8, H1–H5, H10
User of another tenant	any throwaway tenant	Needed for A9 only
Zoho Desk reference	docs/Credentials/ZOHO DESK Credentials.txt (CEO), desk.zoho.com, Edge only , read-only	Works

Never print passwords or tokens in evidence.

1.2 Test data

- Prefix every created module, layout, section, field, picklist value and record with QA MF (e.g. module QA MF Assets / QA MF Asset , field QA MF Serial).
- Module ids: Tickets 88e8ea44-1658-4df3-9930-d0902b91073d · Contacts 43004ce2-577f-4eda-b00a-0750fe40c3d4 · Accounts 784400e1-4a2a-42c2-b4f3-7ca56e79828a · Products 95b45992-5beb-4f23-b619-e3ae0e304933 · Calls f7e488b5-0232-42fd-811c-1f24356af653 · Events 6fb19115-9356-40f0-b064-840d8f1fad93 · Tasks d120a6c2-2596-4254-968b-a1bcd3a000bf · Contracts 0924eb24-e42a-4eab-8efd-e3428f19e882 (Time Entries 6772ffa7-... out of scope).
- Ticket "Standard" layout 1b55c25d-eb27-4046-afd1-b292398b435b (default, all users).
- Write tests on standard modules use a QA MF custom field or a QA MF cloned layout, never the seeded fields, unless the row is about protecting a seeded field.
- A working custom module for groups G and I: create QA MF Assets in C1 and reuse it; delete it at cleanup (needs zero records).

1.3 Baseline (restore point)

Before group A save to baseline/ :

- GET /modules → modules.json (ids, labels, sortOrder, status, rowVersion, recordVisibility).
- For each in-scope module: GET /modules/{id}, /fields, /layouts, /layout-rules, /validation-rules, /dependency-maps, /field-permissions, /related-modules, /links, /buttons.
- GET /fields/datatypes, GET /custom-datatypes, GET /teamspace.
- The Organize Tabs order of each test user (screenshot + the persisted value). Revert = delete every QA MF object, restore orders, and prove it with the final diff (\$Reporting).

1.4 Environment blockers

Blocker	Rows affected	Action
0 departments visible to the owner (GET /desk/departments → [])	C4, C5 (Department storage), Z12, Z13-Department	Inconclusive until a department exists — please add one department and assign the test agents
No lower-profile account confirmed 	every  row	Inconclusive until provided
No custom module on the tenant	C11, K12, L8	Create QA MF Assets in C1
No module with a subform or rollup in use	I33–I36	Create on QA MF Assets
Seeded standard fields were already soft-deleted on 14 Sep 2026 (Contacts: Mailing Address, Description; Accounts: Industry, Fax, Address, Description)	F7, Z11, Z13	Record as found; do not restore without the product owner's OK (another tester may have done it on purpose)
Leftover custom data type test_data_type ("TestDataType", 25 Aug 2026) shows in every palette	J4	Leave it; do not use it in new tests

1.5 Things that cannot be undone cleanly

- **API names are fixed** after a field or module is saved — a QA MF name stays in audit and history.
- **Delete Permanently** of a field removes its data; never do it on a seeded field.
- A module with records cannot be deleted — delete QA MF records first (and purge them from the Recycle Bin, NDC-1637).
- Module Data Storage cannot be changed after create.
- Auto-number counters do not go back.
- Integration-contract warnings: adding/removing fields on a module used by an integration changes its API schema version — check the banner before any write on Tickets/Contacts/Accounts/Products/Contracts (CRM ⇌ Desk sync uses them).

2 Surface and contract

2.1 Admin API (all under `/api/v1`, header **Authorization: Bearer ...**, **x-app-key: desk**)

Method	Path	Use
GET/POST	<code>/modules</code>	list / create module
GET/PATCH/DELETE	<code>/modules/{id}</code>	read / rename / delete
GET/POST/PATCH/DELETE	<code>/modules/{id}/fields[{fid}]</code>	fields
GET	<code>/modules/{id}/fields/{fid}/usage</code> , <code>/value-usage</code> , <code>/unique-peers</code>	delete check, option replacement, unique across modules
GET/POST/PATCH/DELETE	<code>/modules/{id}/layouts[{lid}]</code> , <code>/layouts/accessible</code> , <code>/layouts/{lid}/record-usage</code> , <code>/layouts/assignments[{aid}]</code>	layouts
GET/POST/DELETE	<code>/modules/{id}/layout-rules</code> , <code>/validation-rules</code> , <code>/dependency-maps</code> , <code>/field-permissions (+ /effective)</code> , <code>/links</code> , <code>/buttons</code> , <code>/record-locks</code>	rules and extras (server routes exist; Desk UI hides most tabs)
GET	<code>/modules/{id}/related-modules</code>	lookups pointing at the module
DELETE	<code>/modules/{id}/sections/{sid}</code>	section delete
GET	<code>/fields/datatypes</code> , <code>/fields/datatypes/{key}</code>	catalogue
GET/POST/DELETE	<code>/custom-datatypes[{key}]</code>	custom data types
GET	<code>/teamspace</code> , <code>/teamspace/{id}/modules</code>	teamspace links
Absent (404, 5 Oct 2026)	<code>/modules/tabs</code> , <code>/tabs</code> , <code>/web-tabs</code> , <code>/global-sets</code> , <code>/modules/order</code> , <code>/modules/limits</code> , <code>/audit-log</code> , <code>/related-list-modules</code>	evidence for Z rows
404	<code>/modules/{id}/permissions</code>	the client calls it; record what the Module Permission modal uses instead

Responses use `{success, data, error, meta}` ; writes carry `rowVersion` .

2.2 Field register (limits found in the client — the server must enforce the same)

Item	Limit / rule (build)	Zoho Desk (docs)
Module plural / singular name	required; length not limited in the client — find the server limit (C6)	25 chars in the UI, 50 in the API
Module description	optional	200 (API)
Layout name	required, ≤ 40 , unique in the module	150 (API clone)
Field label	≤ 50 per language (EN/AR/FR)	250 (API <code>displayLabel</code>)
API name	<code>^[A-Za-z][A-Za-z0-9_]*\$</code> , fixed after save	generated, fixed
Custom fields per layout	512 (client "Custom Fields Left")	230 custom fields per module (Enterprise); 400 fields per layout
Single line	<code>max_length</code> 1–20,000	255 default
Multi line	<code>max_length</code> 1–100,000	32,000
Decimal	precision 1–28, scale 0–10	decimal places + rounding
Currency	precision 1–28, scale 0–4, symbol ≤ 4 chars	rounding options
Auto number	prefix/suffix, start 0–2,147,483,647, digits 0–12, increment 1–2,147,483,647, 1 per layout	not available
Pick list / multi-select / radio / status	<code>allowed_values</code> required, optional <code>max_items</code>	500 options for nested
Lookup	target module (or several), display field, related list title unique per target	5 per module (Enterprise) + filters/search/display/autofill
Pipeline	system only, 1 per layout	—
Department, Team	system only	Department mandatory, cannot be removed
Section	1 or 2 columns; Quick Create = 1 section	sections, columns not documented

Capability flags per type (from `/fields/datatypes`): e.g. Formula/Rollup/Auto number cannot be Required; Lookup, Multi-select, Address, File, Image cannot be Unique; layout rules not allowed on most basic types. I-rows check that the UI greys out the same options.

2.3 Page states and messages (en, verbatim from the locale)

- Create module: "Enter the plural module name." · "Enter the singular module name." · "The storage level cannot be changed after the module is created." · "You need at least one teamspace to create a module." · toast "Module {name} created".
- Delete module: "This permanently deletes the module along with its fields and layouts... A module that still has records can't be deleted..." · "{name} still has records. Delete its records first, then delete the module." · "Failed to delete module".
- Disable: "This module has dependencies and cannot be disabled."
- Layouts: "Layout name cannot exceed 40 characters." · "A layout with this name already exists" · "Name is required" · "This is the default layout and cannot be switched off..." · "...cannot be deleted. Set another layout as the default first." · "Only an active layout can be the default. Enable it first." · "There is no other active layout on this module to move them to. Create one first." · "This permanently deletes the layout and its access settings. This action cannot be undone."
- Builder: "Layout saved successfully." · "Cannot save: {fields} cannot be both Required and Read-only..." · "Cannot save: {label} has no subfields defined..." · "Cannot save: {label} has no rollup configuration..." · "Required fields stay on every layout. Turn off Required first." · "Quick Create supports only 1 section" · "Cannot delete: section contains required fields" · "You have not saved your changes." / "Yes, Leave Page" / "Stay Here" · "Reset all changes?" · "Switching layouts will discard your unsaved changes."

- Fields: "Label must be 50 characters or fewer." · "Must start with a letter and contain only letters, digits, or underscores" · "Invalid regular expression" · "A formula field needs an expression before the layout can be saved." · "A rollup needs its related list, field and function before the layout can be saved." · "Select at least one module — until then the record form can't open a record picker for this field." · "Requires the "Manage field permissions" permission."
- Remove field: "Move to Unused Fields" / "Delete Permanently" · "This field is in use and cannot be deleted." · "Remove the field from those first (Settings → Automation / Process Management)."
- Organize Tabs: "Choose which modules appear in the top bar... Changes apply to you only." · "Always shown" · "Reset to default" · "Your tab arrangement could not be saved. It stays applied here until you reload." · "Every module is on the bar."
- Placeholders: Web Tabs "Embed external URLs as tabs inside the CRM..." / Global Sets "...share them across modules..." + "In Development".

2.4 Data classes

D1 empty · D2 whitespace only · D3 single char · D4 max length · D5 max+1 · D6 leading/trailing spaces · D7 case variants (duplicate check) · D8 Arabic · D9 mixed Arabic/Latin + RTL marks · D10 emoji/astrol · D11 HTML/script (`<img src=x onerror=alert(1)>`) · D12 SQL-ish (`' OR 1=1 --`) · D13 path/URL chars (`../ , %2e`) · D14 zero-width/control chars · D15 numeric edge (0, negative, decimal, `1e9, 15,000` , Arabic-Indic `١٥٠٠٠`) · D16 date edge (29 Feb, DST, 23:59 Cairo vs UTC) · D17 type confusion (number as string, array for scalar, object, null) · D18 unknown key / datatype key · D19 duplicate (same name twice, same API name) · D20 limit+1 items.

Group A — Access and permissions

ID	Scenario	Steps	Expected	Tags
A1	Admin sees Customization	Owner → Setup	CUSTOMIZATION shows Modules and Fields, Data Types, Capabilities, Organize Tabs	[func]
A2	Agent without <code>metadata.modules.manage</code>	 agent opens Setup and <code>/settings/modules-and-fields</code> by URL	Menu item hidden; direct URL shows access denied, no module data rendered	[sec] 
A3	API gate for module writes	 agent token: POST <code>/modules</code> , PATCH <code>/modules/{id}</code> , DELETE <code>/modules/{id}</code>	403 each; nothing changed (GET diff)	[sec] 
A4	API gate for field/layout writes	 agent: POST <code>/modules/{id}/fields</code> , PATCH a layout	403; no field/layout created	[sec] 
A5	Read API for agents	 agent: GET <code>/modules</code> , <code>/modules/{id}/fields</code>	Allowed only as needed to render records; record whether hidden/Don't-Show fields leak in metadata	[sec] 
A6	Field permissions page needs its own right	 admin-like profile without <code>metadata.fields.manage_permissions</code> opens Field Permissions	Read-only with "Requires the "Manage field permissions" permission."	[sec] 
A7	Organize Tabs for every user	 agent opens <code>/settings/organize-tabs</code>	Allowed (personal); cannot change other users' bars	[func] 
A8	Module hidden by Access Control	Set QA MF Assets Access Control = Selected Users (not the agent) →  agent: tab bar, More modules, direct record URL, API GET records	Not listed; URL and API refused (compare NDC-1864 for disabled modules)	[sec]  [serial]
A9	Other tenant	Another tenant's token: GET <code>/modules/{NDC module id}</code> , <code>/fields</code>	404/403; nothing returned	[sec]
A10	No / expired / revoked token	GET <code>/modules</code> with no bearer, an old bearer, a revoked one	401 (<code>SESSION_REVOKED</code> for revoked); no data	[sec]

Group B — Module list

ID	Scenario	Steps	Expected	Tags
B1	Renders	Open Modules and Fields	9 rows (Tickets...Time Entries) with Displayed In Tabs As, Module Name, Last Modified, Status; "1–9 of 9"	[func]
B2	Last Modified shows a person	Hover the avatar	Seeded modules show the system user's name, not "UN" (today createdByName:null → "UN" — possible bug)	[ux]
B3	Search	Search "tick", "TICK", Arabic text, no match	Case-insensitive filter; empty state for no match	[func] [data]
B4	Paging	Rows per page 10/25/50/100	Counts and "Page 1 of 1" correct	[func]
B5	Disable a module [serial]	Toggle QA MF Assets off → open its records as admin and agent → toggle on	Disabled module hidden from tabs/create menu and its records cannot be created (NDC-1864 open)	[func]
B6	Disable a module with dependencies	Toggle Tickets off	Refused: "This module has dependencies and cannot be disabled."; status unchanged	[edge]
B7	Placeholders	Open Web Tabs, Global Sets	"In Development" badge; text should say Desk, not "CRM" (today "inside the CRM" — possible bug)	[ux]
B8	Row menu per module	Open the menu on each standard module and on QA MF Assets	Record the items; "Lead Conversion Mapping" must not appear in Desk; every item that shows must open a working page	[func]
B9	Reorder modules [serial]	Change order, reload; check tab bar and Create menu	"Module order saved"; order persists; record whether tab bar follows	[func]
B10	Setup wording	Setup Home	Says Desk, not "Set up your CRM" (possible bug)	[ux]

Group C — Create and delete a custom module [serial]

ID	Scenario	Steps	Expected	Tags
C1	Create (organization)	Create New Module → plural QA MF Assets , singular QA MF Asset , Organization, profiles Desk Administrator + Desk Agent → Create	"Module QA MF Asset(s) created"; row appears; GET shows storageScope:"organization" , appKey:"desk" , generated apiName	[func]
C2	Required names	Leave plural, then singular empty	"Enter the plural module name." / "Enter the singular module name."; no POST	[neg]
C3	Name data classes	D2, D3, D4/D5 (find limit), D6, D8, D10, D11, D14 in plural and singular	Trimmed; script stored as text and rendered escaped everywhere (list, tab bar, breadcrumbs); a clear limit message (Zoho: 25)	[data]
C4	Department storage	Create QA MF Dept with Department	Created with storageScope:"department" ; records are kept per department	[func] (\$1.4)
C5	Storage immutable	Rename QA MF Dept ; try PATCH storageScope by API	Not offered in UI; API refuses or ignores	[edge] (\$1.4)
C6	Duplicate names	Create a second module named QA MF Assets , then qa mf assets , then Tickets	Refused with a clear message (no two tabs with one name)	[neg] [data]
C7	Module Permission empty	Create with no profile ticked	"Leave empty to keep the default visibility" — record who can open it	[func]
C8	Default content	Open the new module's layouts and fields	Standard layout (default) with the record-name field; system fields Created/Modified Time/By, Owner	[func]
C9	No teamspace	(if a tenant without a teamspace exists) open Create	Disabled with "You need at least one teamspace to create a module." else Inconclusive	[edge]
C10	Double submit	Click Create twice fast	One module only	[edge]
C11	New module surfaces	After C1: top bar, More modules, Create menu ("New QA MF Asset"), Organize Tabs, global search, Recycle Bin module filter, Workflow Rules module list, Import/Export	Appears wherever Desk modules appear; record each place	[func]
C12	Delete with records	Add one QA MF record → Delete module	"... still has records. Delete its records first..."; module kept	[neg]
C13	Delete empty module	Delete the record (and purge from Recycle Bin) → Delete module → confirm	Module, fields, layouts gone; tab gone; no orphan records in Recycle Bin (NDC-1637)	[func]
C14	Team Module	Click Team Module	Record what it creates; it must not break Desk tabs	[func]

Group D — Rename a module [serial]

ID	Scenario	Steps	Expected	Tags
D1	Rename custom module	Rename QA MF Assets → QA MF Devices / QA MF Device	Saved; list, tab, breadcrumbs, record pages, "New QA MF Device" use the new names	[func]
D2	Rename standard module	Rename Products → QA MF Items / QA MF Item ; check everywhere; rename back	Zoho: shown in all standard pages except custom reports/dashboards. Restore exactly	[func]
D3	Tab bar uses the new name	During D2 and a Tickets rename (then restore)	Tab bar, More modules, Organize Tabs and Create menu show the renamed label (today tab names come from the locale — check)	[func]
D4	API name change warning	Change API name of QA MF Devices	Warning "Changing the API name may break existing integrations and automations."; workflows referencing it still work or are flagged	[edge]
D5	Name data classes	D1, D2, D5, D8, D11, D19 (name of another module)	Same rules as C2–C6	[data]
D6	Icon and description	Change icon and description	Shown on tab / summary	[func]
D7	Stale write	Tab 1 and tab 2 open Rename on the same module; save tab 1, then tab 2	Tab 2 gets a conflict (rowVersion) or reloads — no silent overwrite	[edge]
D8	Arabic UI	Switch to Arabic, open Rename	Labels translated; RTL layout correct	[i18n]

Group E — Module detail page

ID	Scenario	Steps	Expected	Tags
E1	Tabs on Desk	Open Tickets detail	Today: Layouts, Fields, Workflow Rules, Summary. Layout Rules / Validation Rules / Map Dependency Fields are Zoho features (Z1, Z2) — record if they appear	[func]
E2	Summary values	Summary on each module	Matches GET /modules/{id} ; dates in tenant time zone; "—" for empty	[func]
E3	Direct URL to unknown module	/settings/modules-and-fields/00000000-... and a CRM module id	Not-found state, no crash, no CRM data	[edge] [sec]
E4	Workflow Rules tab	Open on Tickets	"No workflow rules yet" or the list; New Rule opens the rule editor scoped to the module	[func]

Group F — Fields listing and standard fields

ID	Scenario	Steps	Expected	Tags
F1	Listing	Fields → Field Listing on Tickets	Every field with label, API name, Data Type, Custom Field mark, Layouts; system fields marked	[func]
F2	Layout filter	All Layouts / Standard / Not on any layout	Correct subsets; "No fields on this layout." when empty	[func]
F3	Standard vs custom mark	Compare Ticket Subject, Status, Contact, Priority	Zoho standard fields are not custom fields. Today they show the Custom Field tick (<code>isCustomField:true</code>) — Z11	[func]
F4	Standard field set — Tickets	Compare with Z13	Live Zoho: Contact Name* (system-mandatory, non-removable), Department (non-removable), Account Name, Email, Phone, Subject*, Description, Status* (Open/On Hold/Escalated/Closed, default Open), Ticket Owner, Product Name, Resolution (edit mode only), Skills, Due Date, Priority (-None-/High/Medium/Low), Channel, Language, Classifications; Unused: Category, Sub Category; plus Ticket Id (Autonumber) and Layout	[func]
F5	Standard field set — Contacts, Accounts, Products	Compare with \$Z13	Zoho default fields present and active	[func]
F6	Standard field set — Calls, Events, Tasks, Contracts	Compare with \$Z13	Zoho default fields present; required where Zoho requires	[func]
F7	Soft-deleted standard fields	Field Listing on Contacts and Accounts	Today Mailing Address, Description (Contacts) and Industry, Fax, Address, Description (Accounts) are soft-deleted — record, see §1.4	[func]
F8	Picklist value labels	Open record forms for Ticket Priority, Channel, Classification, Language	Human labels (High, Email, Question, English) — not stored keys (<code>low</code> , <code>email</code> , <code>question</code> , <code>english</code> show in the builder preview today)	[ux]
F9	Hidden "Name" field	Tickets/Calls/... "Name" is required but on no layout	Creating a record from UI and API works without it, or the requirement is documented — record	[edge]
F10	Create and Edit Fields	Click the button	Opens the layout builder of the chosen layout	[func]

Group G — Layouts [serial] (on QA MF Assets and a cloned Ticket layout)

ID	Scenario	Steps	Expected	Tags
G1	Standard layout is default	Layouts tab	"Default" badge with tooltip "New records use this layout unless another one is chosen."	[func]
G2	Create layout	Create New Layout QA MF L1	Created, active, not default	[func]
G3	Clone layout	Clone Ticket Standard → QA MF Ticket clone	Same sections/fields; independent afterwards	[func]
G4	Name rules	D1, D5 (41 chars), D19 (existing name), D8, D11	"Name is required" / "...cannot exceed 40 characters." / "A layout with this name already exists"; Arabic kept; script escaped	[data]
G5	Set default	Set QA MF L1 default, then disable it	Default moves; disabling the default refused	[func]
G6	Default from inactive	Disable L1, Set as default	"Only an active layout can be the default. Enable it first."	[neg]
G7	Delete default	Delete the default layout	"...cannot be deleted. Set another layout as the default first."	[neg]
G8	Delete with records	Create 2 records on L1 → delete L1	"Move records to" picker; records moved; GET record shows the new layout	[func]
G9	Delete the only active layout	Module with one active layout	"There is no other active layout... Create one first."	[neg]
G10	Layout permission	Assign L1 to Desk Agent only →  agent creates a record	Agent can pick only allowed layouts; admin sees all	[sec] 
G11	Layout chooser on create	Two active layouts → New record	User chooses a layout; default preselected	[func]
G12	Limit	Create layouts until refused (Zoho: 20 per department)	A clear limit or record "no limit"	[edge]
G13	Views	Switch CREATE / QUICK CREATE / DETAIL VIEW	Each keeps its own arrangement; record forms follow it	[func]
G14	Quick Create one section	Add a 2nd section in Quick Create	"Quick Create supports only 1 section"	[neg]
G15	Preview	Preview as Create / Quick Create / Detail View	Matches the saved form	[func]
G16	Add section	NEW SECTION, rename EN + AR	Shown in record form in both languages	[func] [i18n]
G17	Columns	1 Column ↔ 2 Columns	Fields re-flow; full-width fields stay full width	[func]
G18	Tab order	Left to Right / Top to Bottom	Keyboard Tab moves in that order on the record form (NDC-492 open)	[a11y]
G19	Reorder sections	Drag and Move Up/Down	Order saved	[func]

ID	Scenario	Steps	Expected	Tags
G20	Delete section	Section with optional fields → delete	"All fields inside will be moved to Unused Fields."	[func]
G21	Delete section with required	Section with a required field	Delete disabled: "Cannot delete: section contains required fields"	[neg]
G22	Unsaved changes	Change, click Back / switch layout	"You have not saved your changes." / "Switching layouts will discard..."; Stay keeps changes	[ux]
G23	Reset	Change, Reset	"Reset all changes?"; restored to last save	[func]
G24	Undo/redo and shortcuts	Ctrl+Z / Ctrl+Y / Ctrl+S / Ctrl+Shift+S	Work as listed in Help	[ux]
G25	Double save	Ctrl+S twice fast	One save; rowVersion + 1	[edge]
G26	Stale two-tab save	Tab 1 and tab 2 edit the same layout; save both	Second save refused or merged — no lost change	[edge]
G27	Session expiry mid-edit	Edit, wait for token expiry/revoke, Save	Clear error; edits not lost silently	[edge]
G28	Integration warning	Add a field on Contacts (CRM sync)	Banner and "This change affects integrations" dialog naming the integration	[func]
G29	Module image	Turn module image On/Off	Image field in the record header follows	[func]
G30	Help drawer	Open Help; search	Articles listed; "No articles match your search." for nonsense	[ux]

Group I — Fields in the Layout Builder [serial] (on QA MF Assets)

ID	Scenario	Steps	Expected	Tags
I1	Palette	List New Fields	33 types; system-only types (Department, Team, Pipeline) not addable — "system only" title	[func]
I2	Add by drag and by click	Add a Single Line both ways	Field placed; counter "Custom Fields Left" – 1	[func]
I3	Field limit	Add fields until 0 left	Further adds refused with a message (client limit 512 per layout)	[edge] [perf]
I4	Required	Mark as Required	Record form requires it; API POST without it → 400/422	[func]
I5	Required + read-only	Set both	Save blocked: "Cannot save: ... cannot be both Required and Read-only..."	[neg]
I6	Unique	Mark Unique; create two records with the same value (UI and API), also D7 case and D6 spaces	Second refused; record whether "abc" = "ABC"	[func] [data]
I7	Unique across modules	Same API name on two modules, tick the other module	Value used in one is refused in the other; "Enforced" shown	[func]
I8	Unique not offered	Lookup, Multi-select, Address, File, Image	Mark as Unique disabled	[neg]
I9	Full/half width	Toggle	Record form follows	[ux]
I10	Remove from layout of a required field	Remove a required field	"Required fields stay on every layout. Turn off Required first."	[neg]
I11	Label	D1, D4 (50), D5 (51), D8, D10, D11 in EN/AR/FR	"Label must be 50 characters or fewer."; escaped; shown per UI language	[data] [i18n]
I12	API name rules	1abc , a-b , a b , ا ب ج د , duplicate of another field; then change after save	Error "Must start with a letter..."; duplicate refused; locked after save (NDC-1873 open: label change doesn't update a new field's API name)	[data]
I13	Type change after save	Try to change a saved field's data type (UI and PATCH datatypeKey)	Not allowed (Zoho: "the field type cannot be changed")	[neg]
I14	Single Line length	max_length 1, 20000, 20001, 0	Bounds enforced in UI and API; record value of length max+1 refused	[data]
I15	Multi Line length	100000 / 100001	Same	[data]
I16	Number / Long Integer / Decimal / Percent / Currency	D15 values; min/max; precision/scale; currency symbol 4/5 chars	Stored as typed; out-of-range refused; scale rounding consistent UI vs API	[data]

ID	Scenario	Steps	Expected	Tags
I17	Date / Date Time	D16 values	Stored in UTC, shown in tenant time zone	[data]
I18	Email / Phone / URL	valid, invalid, D11, <code>javascript:alert(1)</code> URL	Invalid refused; URL rendered safe (no <code>javascript: </code>)	[data] [sec]
I19	Checkbox	default on/off	Default applied on create only	[func]
I20	Pick list options	Add, reorder, remove; D1, D19 duplicate value, D8, D11	Duplicates refused; Arabic kept; escaped	[data]
I21	Remove a used option	Option used by 2 records → remove → Save	"Replace removed options" dialog with counts; Replace and apply updates the 2 records; Leave empty clears them	[func]
I22	Multi-select max items	<code>max_items 2</code> , choose 3	Refused	[edge]
I23	Radio / Status	Add options; Status as kanban/pipeline source	Works on record form	[func]
I24	Default value	Default for text, pick list, checkbox	Prefilled on create; not applied to existing records	[func]
I25	Regex	<code>^[A-Z]{2}[0-9]{4}\$</code> ; tester with match/no match; invalid ([	"✓ Value matches..." / "X Value does not match" / "Invalid regular expression"; record form enforces; API enforces	[func]
I26	Regex ReDoS	<code>(a+)+\$</code> with $30 \times \text{"a"} + \text{"b"}$	Server stays responsive (< 2 s)	[sec] [perf]
I27	Conditional visibility	Show field B when A = "X" (case-insensitive), Required while visible	Shown/hidden live; required only while shown; API create obeys the same rule (record)	[func]
I28	No controllers	Layout with one field	"This layout has no other field that could control this one..."	[edge]
I29	Auto-fill	From logged-in user (name/email), from another field, from agent setting	Filled only while empty; never overwrites a saved value (NDC-1231 open)	[func]
I30	Auto number	Prefix QA- , suffix -EG , start 7, digits 4, increment 2; add to a module with records	Preview "QA-0007-EG then QA-0009-EG"; existing records numbered; no duplicates under 5 parallel creates	[func] [edge]
I31	Auto number bounds	digits 13, increment 0, start -1, prefix too long	"Enter a number from ... to ..." / "Must be N characters or fewer."	[data]
I32	Lookup	Lookup to Contacts, display field, related list title QA MF Assets ; second lookup to Contacts with the same title	Related list appears on Contact; duplicate title refused	[func]
I33	Multi-module lookup	"Let users choose the related module...", pick Contacts + Accounts; then none	Record form lets user choose; none → "Select at least one module..."	[func]

ID	Scenario	Steps	Expected	Tags
I34	Formula	quantity * unit_price , functions if , concat , round(x,2) , isblank ; empty expression; unknown field; divide by zero	Computed on save; errors clear; empty → "A formula field needs an expression..." (NDC-1586)	[func] [data]
I35	Subform	Subform with 0 columns → Save; then 3 columns (Single Line, Number, Pick List)	"Cannot save: ... has no subfields defined..."; then rows work on record form (NDC-1335, NDC-1340)	[func]
I36	Rollup summary	Count and Sum of a related module via a lookup; module with no lookup back	Values correct; "The selected module has no lookup field pointing at this module..." only when true (NDC-1444)	[func]
I37	Quick-create trigger	Option "Escalate" opens a Task quick-create	Pop-up opens when the value changes to that option	[func]
I38	File / Image / Camera	Extensions and max size; upload wrong type, too big, SVG with script	Refused with message; SVG not executed	[func] [sec]
I39	Address / Location / Map	Fill subfields; map renders	Stored as object; shown on detail	[func]
I40	Rich text	Paste D11 and a <script>	Sanitised on save and render	[sec]
I41	Remove field → Unused	Remove QA MF Serial → Move to Unused Fields → Save → add back	Data kept and shown again	[func]
I42	Delete permanently	Delete an unused QA MF field	"...cannot be undone."; field gone from list, views, workflow pickers (NDC-1863: 400 on save)	[func]
I43	Delete a used field	Field used in a workflow rule	"This field is in use and cannot be deleted:" with the rule listed	[neg]
I44	Telephony usage	Field used by telephony (if any)	Warning + acknowledge checkbox	[edge]
I45	Delete a seeded standard field	Try on Ticket Subject, Calls Subject, Product Name (UI menu + DELETE API) — do not confirm on seeded fields; stop at the dialog / use a cloned layout	Zoho: standard fields cannot be deleted or moved to Unused. Record which are offered — Z11	[sec]
I46	Unused counter	Remove 3 fields	"Unused Fields 3"; "All fields are in use" when 0	[func]
I47	Field Dependency rule	When Pick list = A → Show B, Require C	Record form follows; Save Rules persists	[func]
I48	Dependency edit/delete	Edit and delete a rule	List updates; empty state text	[func]
I49	Circular dependency	A shows B, B hides A	Refused or harmless; no infinite loop	[edge]
I50	Per-profile permissions	PERMISSIONS tab: agent Read only / no Read	Saves immediately ("...not part of Apply"); new unsaved field shows "Save the layout first..."; enforced	[sec] 

ID	Scenario	Steps	Expected	Tags
			on UI, API and import (NDC-1337, NDC-1593, NDC-1599)	

Group H — Permissions and visibility

ID	Scenario	Steps	Expected	Tags
H1	Field Permissions page	Fields → Field Permissions on Tickets	Profiles × fields grid with Read and Write / Read Only / Don't Show; system fields locked "System field — locked to Read Only"; pagination 25/50/100 (NDC-1336)	[func]
H2	Don't Show	Hide QA MF Serial from Desk Agent	 agent: not on form, list columns, filters, export, API record body	[sec] 
H3	Read Only	Read Only for agent	Visible, not editable in UI; API PATCH refused	[sec] 
H4	Required vs Don't Show	Required field set to Don't Show for a profile	Refused (Zoho: mandatory fields can't be Read Only or Don't Show)	[neg]
H5	Light Agent	Read and Write for Desk Light Agent	Record result (Zoho: Light Agent cannot get Read & Write)	[func]
H6	Record visibility Public	Set Public Read/Write/Delete on QA MF Assets	All users see and edit all records	[func] 
H7	Public Read Only	Switch	Users edit only own and subordinates'	[func] 
H8	Private	Switch	Users see own and subordinates'; unassigned visible to all	[func] 
H9	Access Control	Summary → Selected Users / profiles / org units; empty selection	"Select at least one user, profile, permission, or org unit — or switch back to All Users."	[func]
H10	Module Permission at create	Profiles ticked at C1	Only those profiles see the tab	[sec] 

Group J — Data Types and Capabilities

ID	Scenario	Steps	Expected	Tags
J1	Catalogue	Open Data Types	35 types incl. TestDataType; search "No datatypes match your search"	[func]
J2	Capability flags	Open a type, view flags	Match /fields/datatypes capabilities; builder menu greys out the same actions	[func]
J3	Add custom data type [serial]	QA MF Code on Single Line	"Inherits validation + UI hints"; appears in palette of every module	[func]
J4	Delete custom data type in use	Use J3 type on a field, then delete the type	Refused or the field keeps working — record	[edge]
J5	Capabilities page	Open	"The system catalog seeds 8 capabilities..."; system entries cannot be deleted	[func]
J6	Add / delete capability [serial]	QA MF cap	Created; delete asks to confirm	[func]
J7	Access	 agent opens both pages by URL	Access denied	[sec] 
J8	Data classes	D1, D11, D19 in names	Validation and escaping	[data]

Group K — Organize Tabs and the tab bar

ID	Scenario	Steps	Expected	Tags
K1	Renders	Open Organize Tabs	Selected: Tickets, Knowledge Base, Customers, Analytics, Activities, Chat, Community, Social, Contracts; Unselected empty "Every module is on the bar."	[func]
K2	Hide a tab [serial]	Hide Social	Moves to Unselected; gone from the bar after save; reachable by URL? record	[func]
K3	Reorder	Drag and Move up/down	Bar order follows; persists after reload and sign-out	[func]
K4	Locked tab	Try to hide an "Always shown" module	Not allowed	[neg]
K5	Hide everything	Hide all but the locked one	At least one module stays (Zoho: "At least one module must be selected")	[edge]
K6	Reset	Reset to default	Default order back	[func]
K7	Per user	User A hides Social; user B signs in 	B's bar unchanged — Desk: per user; Zoho: organisation-wide, set by admin (Z8)	[func] 
K8	Save failure	Block the save request (offline)	"Your tab arrangement could not be saved. It stays applied here until you reload."	[edge]
K9	Keyboard	Reorder with keyboard only	Move up/down buttons have names "Move {name} up"; focus visible	[a11y]
K10	Two tabs	Change order in two browser tabs	Last write wins without errors; record	[edge]
K11	More modules	Narrow window; open "More modules"; search	Overflow list; "No module matches your search."	[func]
K12	Custom module tab	After C1	QA MF Assets appears in the bar or More and in Organize Tabs	[func]
K13	Grouped tabs	Customers and Activities	Customers = Contacts + Accounts; Activities = Tasks, Calls, Events — same grouping as live Zoho (bar order there: Tickets, Knowledge Base, Contracts, Customers, Analytics, Activities, Community, Social, Chat, IM)	[func]
K14	Products tab	Look for Products in the bar/More	Live Zoho has no Products tab in the bar either (Products is under Setup → Organization, and can be renamed in Rename Tabs); Desk: Setup → General → Products — not a gap	[func]

Group L — Security, i18n/RTL, accessibility, performance, compatibility

ID	Scenario	Steps	Expected	Tags
L1	Stored XSS in names	D11 in module name, layout name, section name, field label, picklist option, description, related list title, help text	Rendered as text in setup, tab bar, forms, list, kanban, export, emails, audit log	[sec]
L2	Type confusion	POST/PATCH with <code>labels</code> as string, <code>allowed_values</code> as object, <code>required:"yes"</code> , <code>unknown_datatypeKey</code> (D17, D18)	400/422 with a clear error; nothing half-saved	[sec]
L3	Read-only keys	PATCH <code>isSystemField</code> , <code>isCustomField</code> , <code>appKey</code> , <code>moduleKey</code> , <code>createdBy</code> , <code>rowVersion</code>	Ignored or refused	[sec]
L4	Mass assignment on modules	POST module with <code>appKey:"crm"</code> , another <code>teamSpaceId</code>	Refused; module stays in Desk	[sec]
L5	IDOR	Use ids of CRM modules / another tenant in <code>/modules/{id}/fields</code>	403/404	[sec]
L6	Audit log	After C1, D1, G2, I2, I42	Each change logged with who/when/what (Zoho Enterprise logs custom modules)	[func]
L7	Arabic UI	Switch language; open every page	All strings translated, RTL mirrored, drag-and-drop works in RTL	[i18n]
L8	Arabic content	Arabic module, section, field, option names	Saved and shown correctly; sort and search work	[i18n]
L9	Keyboard and screen reader	Builder: add field, move section, open menus	Every control reachable; aria labels (e.g. "Actions for {name}") present	[a11y]
L10	Load time	Open the builder of the largest layout (Tickets) 5×	Median < 3 s; no long tasks > 1 s	[perf]
L11	Big layout	QA MF Assets with 200 fields	Builder usable; save < 5 s	[perf]
L12	Browsers	Chromium, Edge, Firefox, Safari/WebKit; 1366×768 and 1920×1080; tablet width	Builder and tabs usable; drag works	[compat]

E2E journeys

ID	Journey	Steps (chained)	Expected
E2E-1	Admin builds a custom module	C1 → add Single Line (unique), Pick list, Lookup to Contacts, Auto number → set permissions → create 3 records → open a Contact and see the related list	Works end to end; related list shows the 3 records
E2E-2	Tailor the ticket form	Clone Ticket layout → add section + QA MF fields → field dependency → assign to Desk Agent →  agent creates a ticket with that layout	Agent sees only that layout's form; rules applied
E2E-3	Rename and reorganise	Rename Products → tab bar, Organize Tabs, Create menu, records → rename back	Names consistent everywhere; restored
E2E-4	Field lifecycle	Add field → fill on records → remove to Unused → add back (data kept) → change options with replacement → delete permanently	Each step as specified; no orphan values; audit entries
E2E-5	Negative recovery	Save with Required+Read-only conflict → fix → stale second tab save → reload → delete a used field (blocked) → remove usage → delete	Every failure has a clear message; final state consistent
E2E-6	Arabic admin	Arabic UI: create module, section and field with Arabic labels; agent in Arabic creates a record	Works in RTL; labels in Arabic

Group Z — Gaps (run to confirm they are still missing or now built)

Z1–Z6 are in NDC-1879 (items 1–6). Z7 is NDC-1866 item 16. **Z8–Z19 are new** (missing list items 1–12) and are not filed in Plane (user asked for no Plane changes). "Live" = seen in the Zoho Desk org on 5 Oct 2026 (Enterprise trial, read-only); "Docs" = Zoho help/API only. Every Z row is expected to fail until the function is built.

ID	Gap	How to test	Expected (Zoho)
Z1	Layout rules (NDC-1879 item 1)	Look for a Layout Rules tab / builder "Create Layout Rule"; create a rule that shows a field and makes another mandatory	Live: Layouts and Fields → Layout Rules, "Create Rule" per module and department. Docs: Show Fields / Show Sections / Set Mandatory Fields; 10 per layout, 50 per department, 25 conditions; manual create/edit only
Z2	Validation rules (NDC-1879 item 2)	Validation Rules tab; rule across two fields with an alert message	Live: Validation Rules page with "Create Rule". Docs: own alert message, run in creation order; bypassed by import, workflow, API
Z3	Picklist values per layout (NDC-1879 item 3)	Two layouts, different Priority values	Each layout has its own values (live: the Tickets "Sales" layout differs from the default one)
Z4	Help Center access per field (NDC-1879 item 4)	Field properties	Live: "Hide from Help Center" toggle and "Editable for End Users ▼" in Edit Properties; help-center icon on each ticket field
Z5	Nested and colour-coded picklists (NDC-1879 item 5)	Pick list properties and palette	Live: "Nested picklist" toggle (6 levels, ::); palette has Colored Picklist and Colored Multiselect
Z6	Field encryption / ePHI (NDC-1879 item 6)	Field properties	Live: "Encrypted Fields 10" in Custom Fields Left. Docs: encrypt a custom field; ePHI label
Z7	Ticket number format (NDC-1866 item 16)	Ticket Number field	Live: "Ticket Id" is an Autonumber field. Desk: read-only Single Line
Z8	Tab bar set by the admin for everyone (item 1)	K7	Live: Setup → Modules and Tabs → Organize Tabs (Selected / Unselected Modules, drag to reorder). Docs: a hidden module "won't appear regardless of user profile"; at least one module must stay. Desk: Organize Tabs is personal ("Changes apply to you only")
Z9	Rename any tab (item 2)	Look for a Rename Tabs page; try to rename Knowledge Base, Customers, Activities, Analytics	Live: Rename Tabs with 19 display names — Tickets, Knowledge Base, Contracts, Accounts, Contacts, Products, Reports, Dashboards, Activities, Time Entry, Community, Social, Chat, Calls, Tasks, Events, IM, Analytics, Customers. Desk can rename only record modules, from the module list
Z10	Search fields per module (item 3)	Look for a "Search Fields" setting	Live: Layouts and Fields → Search Fields → All Fields / Specific Fields (27 ticket fields listed). Docs: up to 10 per module (6 for Contracts, Products, Activities)
Z11	Standard fields are protected (item 4)	F3, I10, I45 (stop at the dialog on seeded fields)	Live: core fields carry "Non-removable standard field" (Tickets Subject, Status, Contact Name, Email, Description, Due Date, Priority, Channel, Ticket Owner; Calls/Events/Tasks Subject; Product Name; Contract Name, Account Name, Start Date); system-mandatory fields cannot be un-required; other standard fields can only go to Unused Fields , never be deleted; no standard field is marked custom. Desk: seeded fields are marked custom; Calls/Events/Tasks Subject, Product Name and Contract

ID	Gap	How to test	Expected (Zoho)
			Name are deletable; Ticket Subject and Status can be made optional (then removed) and Contact, Department, Email, Description, Due Date, Priority, Channel, Ticket Owner, Resolution can be removed from the ticket layout (API flags <code>canMarkRequired</code> / <code>canRemoveFromLayout</code>)
Z12	Department-specific layouts and the Department field (item 5)	G2 with a department (§1.4: needs a department)	Live: Tickets, Tasks, Calls, Events, Contracts have one layout per department (department picker on Layouts); Department is non-removable on all of them and mandatory on Tasks, Calls, Events and Contracts; Products has a mandatory multi-select Department. Desk: layouts are per profile only; Department is optional and not on the Tasks/Calls/Events layouts
Z13	Missing or different standard fields (item 6)	F4–F6	Live: Tickets Category and Sub Category (with a Category → Sub Category dependency). Accounts Annual Revenue (Currency), Industry, Fax, Street, City, State, Code, Description. Contacts Street, City, State, Zip, Country, Description. Products Department, Manufacturer (Pick List), Unit Price. Tasks Remind At ; Calls/Events Remind me . Calls: Contact Name and Call Status mandatory. Events: Contact Name mandatory. Contracts: Account Name , Start Date and Support Plan mandatory, plus SLA Name. Zoho keeps several of these in Unused Fields by default; the gap is that Desk does not have them as standard fields, or has them soft-deleted
Z14	Picklist tools (item 7)	Pick list ADVANCED tab and field menu	Live: Pick List Values editor with import, clear, sort and expand icons and "Add Values in Bulk" ; Replace Values in the field menu. Docs: Replace Values updates existing records at any time; export values as CSV. Desk: add/reorder/remove one by one; replacement offered only when removing a used option
Z15	Rounding options (item 8) — Docs	Decimal / Currency properties	Normal / Round off / Round down / Round up, and decimal places
Z16	Lookup options (item 9) — Docs	Lookup ADVANCED tab	Filter lookup records (5 criteria); search by up to 6 fields; up to 6 display fields in the pop-up; sort; auto-fill up to 5 layout fields from the chosen record
Z17	Layout Help Center settings (item 10)	Create New Layout form and Layouts list	Live: the Tickets Layouts list has a Display in Help Center column. Docs: Add Layout has Display Name in Help Center, Description, "Display in Help Center" and "Allow non-department agents"
Z18	Ticket Status page (item 11)	Look for a status setup page	Live: Layouts and Fields → Ticket Status per department: Add Status, Status Type (Open / On Hold / Closed), Fall-back to default; "Pause the SLA clock with the new On Hold State". Desk: Status is a field with a value list only. Check first whether a Tickets or SLA gap item already covers it
Z19	Agents as a customisable module (item 12)	Look for Agents in Modules and Fields	Live: Agents is a module with its own layout (2 sections), 15 field types and 240 custom fields left. Desk: no Agents module. Check NDC-1878 (Agents gaps) first

Traceability

There is no PRD with numbered requirements for this feature in Plane (NDC-1249 holds test cases; NDC-1489 is a plan row). Traceability is to the Zoho capability areas and existing defects.

Area	Scenarios	Related Plane items
Module list, enable/disable, order	B1–B10	NDC-1864
Create / delete custom module	C1–C14, A3	NDC-1693, NDC-1630, NDC-1637
Rename module (singular/plural, API name)	D1–D8, E2E-3	—
Module detail and rules tabs	E1–E4, Z1, Z2	NDC-1489, NDC-1879
Fields listing, standard fields, search fields, status, agents	F1–F10, Z10, Z11, Z13, Z18, Z19	NDC-1339, NDC-1338, NDC-1598
Layouts	G1–G30, Z3, Z12, Z17	NDC-1588, NDC-492
Field types and properties	I1–I50, Z4–Z7, Z14–Z16	NDC-1873, NDC-1863, NDC-1586, NDC-1444, NDC-1335, NDC-1340, NDC-1231, NDC-1230
Field permissions, record visibility, access control	H1–H10, A5, A6, A8, I50	NDC-1337, NDC-1336, NDC-1593, NDC-1599, NDC-1835
Data types and capabilities	J1–J8	NDC-1445
Tabs	K1–K14, Z8, Z9	—
Security	A1–A10, L1–L5	—
i18n, a11y, perf, compat	L6–L12, D8	—

Every scenario maps to at least one area; every Z row maps to a missing-list item.

Reporting

- Folder: `executions/<run>/` with `report.md`, API request/response (redacted) and screenshots.
- Evidence rule (user, 4 Oct 2026): **screenshots only for failed rows**, each annotated side by side against live Zoho Desk (TAVI left, Zoho right, table "Behaviour | Zoho | TAVI | Match"). Passed rows get a verdict and one line of text.
- Redaction: never write tokens, cookies or passwords — write `Bearer <len=1263>`. Mask other users' emails.
- Each Fail: steps, expected, actual, evidence, and the existing NDC item it matches (see Traceability) before suggesting anything new — run the false-positive checklist (permission, layout scope, cached SPA, stale/revoked token, another tester's change).
- Coverage gaps restated at the end: 🚫 rows and Department-storage rows are Inconclusive while §1.4 blockers stand.
- Revert proof: after cleanup repeat every §1.3 GET and diff against the baseline. Only expected differences allowed (Recycle Bin entries if not purged, auto-number counters), listed by id.